Multi-View Mask R-CNN with Crop Refinement for Panoptic Solar Filament Segmentation

A Solution to the Solar Filament Segmentation Challenge 2026

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Abstract

This report describes our solution to the Solar Filament Segmentation Challenge 2026, a Kaggle competition on automatic segmentation of solar filaments in GONG H-α observations. We use a single Mask R-CNN detector trained on all annotation categories, four-view test-time augmentation with cross-view agreement voting, and a small U-Net that re-segments every detection from a native-resolution crop. On three held-out cross-validation folds the pipeline reaches a Panoptic Quality (PQ) of 0.442 ± 0.014 under our re-implementation of the official per-annotator metric, and 0.39 on the public leaderboard. Most of the gain over our previous submission came from two simple changes: learning the annotated "unidentifiable" filaments and raising the detection-score threshold. Several intuitive additions (fragment merging, faint-filament oversampling, morphological boundary correction, learned candidate rescoring, model ensembling) did not help, and we report them as negative results.

1 Introduction

This report goes over the details of our submitted solution to the Solar Filament Segmentation Challenge 2026 [2]. The challenge, launched on July 10, 2026, is organized by the Earth-Space AI Research (ESAIR) Lab[1]. This challenge utilizes the MAGFiLO dataset for evaluation of the segmentation algorithms [1]. Each filament must be returned as a separate mask, and the Panoptic Quality (PQ) metric [4] penalizes both missed and spurious instances, as well as fragmented or merged ones. We therefore treat the task as instance segmentation rather than semantic segmentation. Our solution uses only the competition training images and annotations, and only the provided test images at inference time; no external data and no leaked labels are used. Code, notebook and trained weights are available at https://github.com/sivapnlr80-cpu/Solar-Filament-Segment.

2 Methodology

Figure 1 summarizes the end-to-end pipeline, which is a single model followed by deterministic post-processing.

Data and splits. We use the 668 annotated training observations of MAGFiLO. Several annotators outlined the filaments of the same image independently and they frequently disagree; we treat every annotator–image pair as its own ground-truth entry, as the organizers' evaluator does. For validation we use 5-fold cross-validation (shuffled, seed 42) over image files; images carrying only "unidentifiable" annotations are used for training only. All reported held-out numbers come from folds on which neither the detector nor the refiner was trained (the refiner is re-trained per fold).

Detector. We fine-tune the torchvision Mask R-CNN with a ResNet-50-FPN v2 backbone [3], initialized from COCO weights, with one foreground class. Images are resized from 2048 to 1024 pixels. The first-round models used only the left/right chirality categories; the final model is trained on *all four* annotation categories collapsed into one "filament" class, because the "unidentifiable" category (37% of all annotations) is scored by the official ground truth but was never learned. Training: batch size 2 with gradient accumulation 4, AdamW (lr 10^{-4}, weight decay 10^{-4}), cosine schedule, 10 epochs, random horizontal/vertical flips; the checkpoint with the best held-out Dice is kept. The submitted model is the fold-0 checkpoint (trained on 80% of the data).

Crop refiner. The 1024-pixel detector masks are coarse. For each detection we take the coarse mask (probability > 0.3 at native resolution), cut a square native-resolution crop around it (side $1.8\times$ the mask extent, at least 96 px), resize it to 256×256 and re-segment it with a U-Net [5] with an ImageNet-initialized ResNet-18 encoder, then threshold at 0.5. The refiner is trained on native-resolution crops around annotated filaments with a $0.5\,\text{BCE} + 0.5\,\text{Dice}$ loss, AdamW (lr 3×10^{-4}), 15 epochs, rotation/flip/gamma/brightness augmentation.

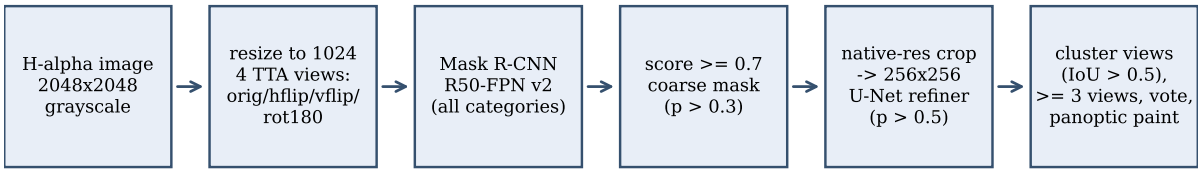
Test-time agreement and post-processing. Each image is processed in four views (original, horizontal flip, vertical flip, 180° rotation). Detections with score < 0.7 are discarded; the remaining refined masks are clustered across views (IoU > 0.5), a cluster survives only if at least *three of the four views* contain it, and its mask is the pixel-wise majority vote of its members. Surviving filaments are painted in descending score order so that masks never overlap, masks smaller than 200 px are dropped, and each mask is run-length encoded. The two thresholds were chosen by cross-validation (Section 3); a candidate is worth emitting when more than roughly 30% of the annotators would match it, which is why a stricter-than-usual score threshold helps under per-annotator PQ.

3 Evaluation

Strategy. We re-implemented the official metric: PQ is computed per annotator–image entry against *all* predictions of that image (a prediction is a true positive for an entry if IoU > 0.5 with one of its filaments; unmatched predictions count as false positives and unmatched ground-truth filaments as false negatives for that entry). The inter-annotator ceiling under this metric, obtained by

[1] www.esairlab.com/

Kaggle Competition, kaggle.com/competitions/filament-segmentation-2026
2026.



Output: one RLE mask per filament, no overlaps, min. area 200 px (single model, fold-0 checkpoint)

Figure 1: Inference pipeline. All images are 8-bit grayscale; the detector sees a 1024×1024 resize (replicated to three channels), the refiner sees native-resolution crops.

Table 1: Held-out PQ per cross-validation fold (4-view agreement ≥ 3, refiner on). Mean ± sample s.d. over three folds; "cat. 1–2" is our earlier detector trained on left/right categories only.

Detector	Score thr.	Fold 0	Fold 1	Fold 2	Mean ± s.d.
cat. 1–2	0.4	0.408	0.428	0.438	0.425 ± 0.015
all categories	0.4	0.409	0.417	0.419	0.415 ± 0.005
all categories	0.5	0.418	0.428	0.432	0.426 ± 0.007
all categories	**0.7**	0.427	0.448	0.452	**0.442 ± 0.014**
all categories	0.8	0.434	0.450	0.447	0.444 ± 0.009
all categories	0.9	0.413	0.427	0.396	0.412 ± 0.015

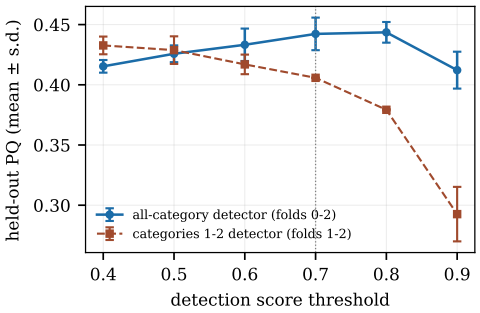


Figure 2: Held-out PQ versus detection-score threshold (error bars: s.d. over folds).

Table 2: Ablations. Reference and variant PQ on held-out data (fold 0 unless stated; all-category detector).

Change	Ref.	Variant
No crop refiner (thr 0.5)	0.418	0.389
Single view instead of ≥ 3 of 4 views (thr 0.5)	0.418	0.390
Fragment merging, gap ≤ 8 / 16 / 32 px (thr 0.4)	0.409	0.409 / 0.397 / 0.360
Erode / dilate masks by 1 px (thr 0.5)	0.418	0.384 / 0.398
Oversample faint-filament images (thr 0.5)	0.418	0.403
LightGBM cluster rescoring (folds 0–2, pooled)	0.441	0.435
Refiner on consensus soft targets (folds 0–2, mean)	0.444	0.447

scoring each annotator against the others, is only 0.34, so a PQ above 0.4 exceeds the PQ between annotators. In our experience cross-validation gains transferred to the public leaderboard (+0.018 in cross-validation, +0.02 on the leaderboard for the final change), but absolute leaderboard values are lower than cross-validation values (0.39 versus 0.44); we did not determine why. Because the two thresholds below were selected on the same three folds that we report, the cross-validation numbers are slightly optimistic.

Main results. Table 1 and Figure 2 show that the all-category detector with a score threshold of 0.7–0.8 is better than our earlier detector at its best threshold in every fold (paired gain +0.018±0.003 at 0.7). The earlier detector degrades quickly when the threshold is raised, whereas the all-category detector keeps a broad optimum between 0.7 and 0.8. We submit threshold 0.7, the safer side of the plateau. On the public leaderboard (about 50% of the 180 test images) the successive versions scored 0.33 (five-fold ensemble, no TTA), 0.36 (+ refiner), 0.37 (single model) and **0.39** (final).

Ablations and negative results. Table 2 lists what each component contributes and what did *not* help. The refiner and the multi-view agreement each add about +0.03. Fragment merging by bridging nearby masks, morphological dilation/erosion, oversampling images with faint filaments (the faintest quartile has the lowest recall), a LightGBM model that rescored detection clusters from score, area, contrast, position and agreement features, and a five-fold ensemble all failed to beat the simple pipeline; the ensemble was slower (about 6×) and scored slightly lower on the leaderboard (0.36 versus 0.37). The one change that did help is small: re-training the refiner on the mean mask of all annotators who drew a filament (soft targets) raised PQ by +0.003 on average and was best in all three folds, while using every annotator's hard masks alone gave no gain; the effect is about as large as the run-to-run noise of the refiner (≈ 0.002), so it is not part of the submitted model.

Distributions. On the fold-0 held-out images, the mean per-image Dice of the union of predicted masks against each annotator is 0.647 (median 0.687). Matched filaments have a mean IoU of 0.68 (Figure 3). Fragmentation is limited: 2.6% of ground-truth filaments overlap

two or more predictions (one-to-many), and 2.8% of predictions overlap two or more ground-truth filaments of the same annotator (many-to-one).

Qualitative results. Figure 4 shows held-out predictions: a good, a middling and a failed image. Failures are dominated by faint filaments that annotators drew but the model scores low, and by extent disagreements (the prediction is a plausible but differently sized outline).

Strengths, weaknesses and efficiency. The pipeline is simple, needs a single checkpoint, and runs on the 180 test images in about 10 minutes on one T4 GPU (428 s for detection, TTA and refinement on the GPU, 158 s for CPU clustering and painting). Its main weaknesses are (i) low contrast: held-out recall rises from 0.35 for the faintest quartile of ground-truth filaments to 0.81 for the darkest (0.62 overall); (ii) annotator disagreement: filaments drawn by one annotator only are recalled at 0.52 versus 0.90 for filaments drawn by three, so a large share of misses is not a model error; (iii) boundary extent: about half of the misses (368 of 697 in fold 0) are localization misses (best IoU between 0.1 and 0.5) whose predicted area is smaller than the ground truth about as often as it is larger (49% versus 43% at threshold 0.5), so a systematic bias cannot be removed by morphological post-processing; and (iv) fine structure such as barbs, which we did not evaluate separately and which the 1024-pixel detector input and 256-pixel refiner crops may limit. Promising directions are a higher-resolution or higher-capacity mask head, soft consensus targets for the detector as well as the refiner, and agreement-aware training losses.

4 Acknowledgment

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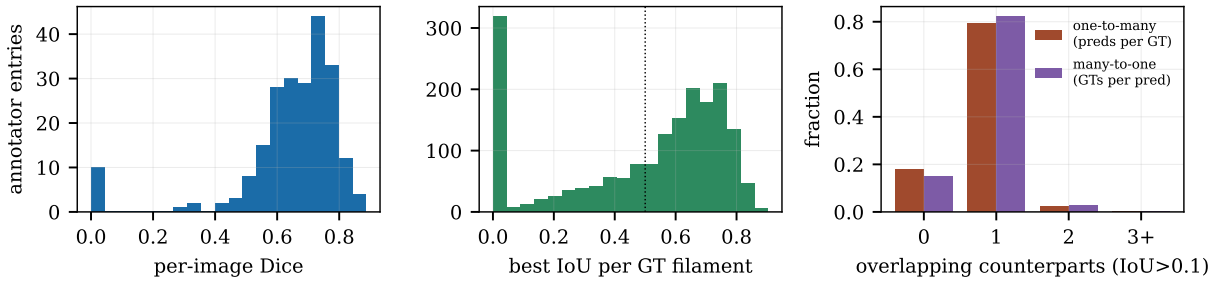


Figure 3: Fold-0 held-out distributions for the submitted configuration: per-image Dice, best IoU per ground-truth filament, and one-to-many / many-to-one relations.

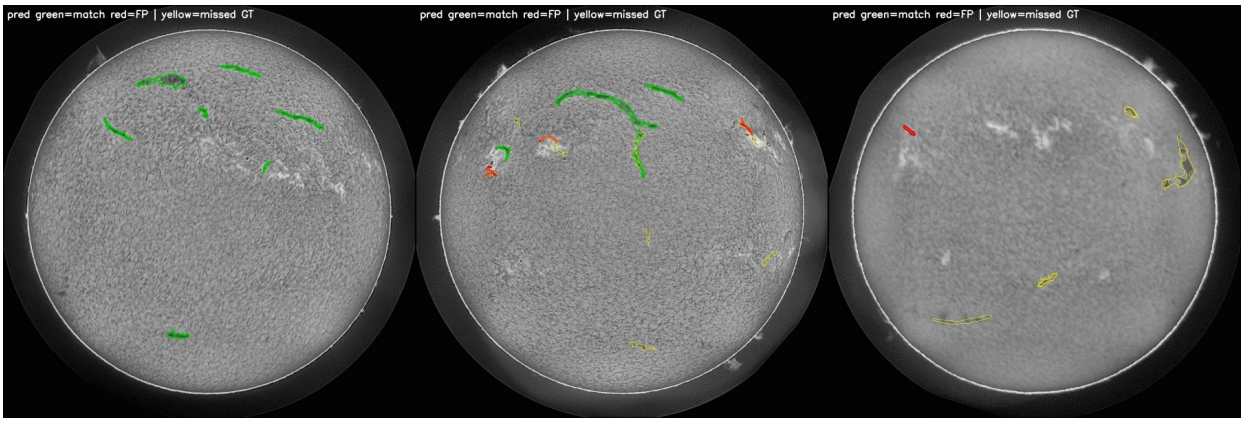


Figure 4: Held-out examples (green: predictions matched to an annotator; red: unmatched predictions; yellow: ground-truth filaments without a match).